AI-Driven Online Proctoring & Anti-Cheat System

Software Requirements Specification Version 01.00

Revision History

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1. Introduction

1.1 Purpose

This document specifies the software requirements for an AI-driven remote exam monitoring platform. It defines the functional and non-functional requirements needed by developers, QA, and stakeholders to design, build, and evaluate the system.

1.2 Scope

This document defines the complete set of functional and non-functional requirements for the AI-Driven Online Proctoring & Anti-Cheat System. It covers requirements for image processing, machine learning-based detection, behavioral and session monitoring, the data pipeline and storage layer, AI-based summary generation, user and account management, examination management, the online examination student experience, the core exam data model, and examination/proctoring reporting, along with the associated usability, reliability, performance, security, design-constraint, and interface requirements detailed in Section 3. It serves as the agreed basis between the development team and project stakeholders for designing, building, testing, and evaluating the system, and is the reference against which the Software Design Document (SDD) and subsequent test plans are traced.

This document does not cover system architecture or design decisions, which are documented separately in the Software Design Document, or project scheduling, staffing, and budget, which are documented in the Software Project Management Plan. It likewise does not specify the authoring of exam question content/item banks, the grading or scoring of exam responses, or identity verification at account-creation time, since these fall outside the boundaries of the system this document describes.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
CV	Computer Vision
ML	Machine Learning
LLM	Large Language Model
FERPA	Family Educational Rights and Privacy Act

GDPR	General Data Protection Regulation
Gaze Vector	Estimated direction of a person's visual attention
Pitch/Yaw/Roll	The three rotational axes used to describe head orientation
Risk Score	A composite numeric indicator of a test-taker's likelihood of cheating
Multimodal Fusion	Combining signals from multiple input types (e.g., audio + video) to make a single decision
Session Summary	A natural-language, LLM-generated recap of a test-taker's flagged events during a session

1.4 References

- FERPA (20 U.S.C. § 1232g)
- GDPR (Regulation (EU) 2016/679)
- [Internal design notes / prior discussion on system architecture]

1.5 Overview

Section 2 describes the system at a high level. Section 3 details specific functional requirements (image processing, machine learning, behavioral & session monitoring, data pipeline, AI summary generation, user & account management, examination management, online examination delivery, the core exam data model, and examination & proctoring reports) and non-functional requirements (usability, reliability, performance, security, design constraints, and interfaces).

2. Overall Description

The system is an AI-driven remote exam monitoring platform combining computer vision and machine learning to detect cheating in real time while minimizing false positives, with human proctors reviewing flagged cases. It operates as a browser- or client-based application that runs alongside an exam session, continuously analyzing webcam video, microphone audio, and browser-level behavioral events. Detected anomalies are scored via an event-driven data pipeline and surfaced to proctors on a real-time dashboard rather than triggering automatic penalties, preserving human judgment in the loop.

Post-session (or on demand), an LLM-based summarization service reads the structured event log for a given session and generates a plain-language recap for the proctor, so that a session with dozens of raw flags can be reviewed as a short narrative rather than a log dump. The summary is advisory only and never itself constitutes a finding.

The system is designed for use by educational institutions and certification bodies administering highstakes remote exams, and must operate within strict privacy and biometric-data regulations.

Beyond real-time monitoring, the platform manages the surrounding examination lifecycle: institutions and instructors provision accounts and rosters, configure and schedule exams with an associated proctoring policy, and deliver the exam itself to students through an integrated online exam experience. After each exam, structured event data, AI-generated summaries, and session recordings feed into reporting and analytics used by instructors, administrators, and appeals reviewers.

3. Specific Requirements

3.1 Functionality

3.1.1 Image Processing (Control Accuracy)

- 3D Facial Landmark Fitting & Head-Pose Estimation: The system shall compute pitch, yaw, and roll from facial landmarks to detect when a student turns away from the screen.
- Gaze Vector & Pupil-Shift Tracking: The system shall track gaze direction and pupil position to flag sustained off-screen looking (e.g., reading hidden notes).
- Baseline Calibration & Drift Detection: The system shall function as a feedback control loop — establishing a "normal" head/gaze position during a calibration step, then measuring angular deviation from that baseline. Natural, brief movement shall be tolerated; sustained drift beyond a configurable threshold shall be flagged.

3.1.2 Machine Learning (Data Accuracy)

- Acoustic Classification: The system shall classify audio input to distinguish benign ambient noise (e.g., traffic, typing) from suspicious sounds (e.g., whispering, a second voice).
- Visual Object/Person Classification: The system shall detect unauthorized objects (phones, notes) or the presence of a second person in the camera frame.
- Multimodal Fusion: The system shall combine audio and video signals to reduce false positives, rather than flagging on a single-modality signal alone.
- Bias Mitigation: Models shall be trained on diverse data to avoid bias across accents, lighting conditions, and home environments.
- Confidence Thresholding & Human Review: All ML-based flags shall pass through configurable confidence thresholds and require human proctor review before any accusation or disciplinary action is finalized — the system shall never issue fully automated accusations.

3.1.3 Behavioral & Session Monitoring

- Behavioral Signal Tracking: The system shall monitor browser-level behavioral signals, including tabswitching, copy-paste actions, and loss of browser focus.
- Encrypted Session Recording: The system shall record exam sessions (audio/video/screen) in encrypted form to support audit trails and appeals.
- Real-Time Risk-Score Dashboard: The system shall provide proctors with a real-time dashboard displaying composite risk scores across all monitored test-takers, enabling efficient monitoring at scale.

Proctoring Event Catalog

Event Type	**Trigger Signal(s)**	**Default Severity**	**Flag Description**
Gaze Deviation	Sustained off-screen gaze/pupil shift beyond threshold	Medium	Test-taker's gaze was directed away from the screen for an extended period.
Head-Pose Deviation	Head pitch/yaw/roll drift beyond calibrated baseline	Medium	Test-taker's head orientation deviated from the established baseline.
Second Voice Detected	Acoustic classification identifies a non-testtaker voice	High	Audio consistent with a second speaker was detected during the session.
Unauthorized Object Detected	Visual object classification (phone, notes, textbook)	High	An unauthorized object was visible in the camera frame.
Second Person Detected	Visual person classification identifies more than one face/body	Critical	More than one person was detected in the camera frame.

Browser Focus Loss / Tab Switch	Browser-level behavioral signal	Medium	Test-taker navigated away from the exam window or switched browser tabs.
Copy-Paste Action	Browser-level behavioral signal	Low	A copy or paste action was performed within the exam window.
Session Connectivity Interruption	Reliability/heartbeat signal	Low	The exam session experienced a temporary loss of audio, video, or network connectivity.

Instructor Event Review Actions

- Acknowledge & Dismiss: The proctor may mark a flagged event as reviewed and non-actionable, retaining the original event in the audit trail.
- Escalate to Academic Integrity Case: The proctor may forward a flagged event, along with its supporting recording reference and risk score context, into a formal academic integrity review case.
- Add Reviewer Note: The proctor may attach a free-text note to a flagged event to record contextual judgment for later reference or appeal.
- Override Severity: The proctor may adjust an event's auto-assigned severity level, with the override and the original system-assigned value both retained for audit purposes.
- Bulk-Review Low-Severity Flags: The proctor may review and dismiss multiple low-severity flags of the same type within a session in a single action, rather than one at a time.

3.1.4 Data Pipeline & Storage

- Event Ingestion: The system shall accept behavioral events and ML flags from the client via a REST/WebSocket ingestion API and place them on a message queue (e.g., Kafka/Kinesis) to absorb load spikes across concurrent exam sessions.
- Stream Processing: A stream processor shall consume queued events, perform multimodal fusion (per 3.1.2), compute the composite risk score, and persist results.
- Structured Storage: The system shall write structured events to an Events/Flags table (session_id, student_id, timestamp, signal_type/error_type, confidence, duration) and risk scores to a RiskScores table (session_id, running score, last_updated).
- Media Storage: Raw audio/video recordings shall be written to encrypted object storage, with only a reference/pointer stored in a Recordings table — never the media itself in the relational database.

- Retention Enforcement: A scheduled job shall purge data from all tables and object storage according to configured retention periods (see 3.6).

3.1.5 AI Summary Generation

- Trigger Conditions: The system shall generate a session summary either automatically upon exam completion (batch) or on demand when a proctor requests one for an in-progress or completed session.
- Data Source: The summarizer shall read only structured rows from the Events/Flags table (student_id, timestamp, signal_type/error_type, confidence, duration) for the relevant session_id. It shall not send raw video, audio, or biometric feature data to the LLM.
- Chronological, Timestamped Narrative: The generated summary shall present each student's flagged events in chronological order, with every referenced event annotated with its original timestamp, so a proctor can cross-reference the narrative directly against the raw Events/Flags log.
- Summary Storage: Generated summaries shall be stored in a SessionSummaries table (session_id, summary_text, generated_at, model_version), linked to the originating session.
- Dashboard Presentation: The proctor dashboard shall display the AI summary as a distinct, clearly labeled panel alongside — never in place of — the underlying raw flags and recordings.
- Advisory-Only Constraint: A generated summary shall never be treated as a finding in itself; it exists to help a proctor triage faster, not to replace review of the underlying evidence (see 3.6).

3.1.6 User & Account Management

- Role-Based Accounts: The system shall support distinct Student, Instructor/Proctor, Administrator, and Appeals Reviewer roles, each with its own permission set.
- Institution/Tenant Provisioning: Administrators shall be able to provision institution-level tenant accounts and bulk-import student and instructor rosters (e.g., via CSV or SIS integration).
- Authentication & SSO: The system shall support username/password authentication as well as institutional single sign-on (SAML/OAuth) for supported identity providers.
- Profile & Consent Management: Users shall be able to manage their profile information, and students shall be able to review, grant, and revoke biometric monitoring consent in line with applicable regulation (see 1.4).

3.1.7 Examination Management

- Exam Creation & Configuration: Instructors and administrators shall create exam records specifying the scheduled window, duration, allowed materials, and the proctoring policy (monitored event types and severities) to apply.
- Roster Assignment & Accommodations: Exams shall be assignable to a class or student roster, with support for individual scheduling accommodations such as extended time or an alternate exam window.
- Proctoring Policy Templates: The system shall support reusable proctoring policy templates (e.g., strict vs. lenient monitoring profiles) that can be applied to one or more exams.

- Exam Lifecycle States: An exam record shall progress through Draft, Scheduled, In Progress, Completed, and Archived states, with each state gating the actions available to instructors and students.

3.1.8 Online Examination (Student Experience)

- Pre-Exam Check-In: Before an exam begins, the student shall complete an identity/device check, camera and microphone calibration, and acknowledgment of monitoring consent.
- Exam Session Interface: The student-facing interface shall present exam content and a timer while proctoring signal capture runs unobtrusively in the background.
- Real-Time Integrity Notices: The system may surface non-punitive, on-screen notices to the student when a recoverable issue is detected (e.g., “face not visible”), giving the student a chance to selfcorrect.
- Technical Support Escalation: A student shall be able to flag a technical issue to a live support/proctor contact without pausing or invalidating ongoing proctoring for the session.

3.1.9 Core Exam Data Model

Entity	Key Fields	Description
Exam	exam_id, title, scheduled_window, duration, policy_template_id	Defines a schedulable exam and the proctoring policy applied to it.
ExamSession	session_id, exam_id, student_id, start_time, end_time, status	A single student’s attempt at an exam; the anchor record that Events/Flags, RiskScores, and Recordings (3.1.4) attach to.
UserAccount	user_id, role, institution_id, auth_provider	A student, instructor/proctor, administrator, or appeals reviewer account and its institutional affiliation.
ProctoringPolicy	policy_id, event_thresholds, severity_map	A reusable configuration of monitored event types, thresholds, and default severities (see 3.1.7).

- These entities extend, and reference by session_id, the Events/Flags, RiskScores, Recordings, and SessionSummaries tables defined in 3.1.4.

3.1.10 Examination & Proctoring Reports

- Session Integrity Report: The system shall generate an exportable report (PDF/CSV) summarizing flagged events, the final risk score, and the AI-generated summary for a single exam session.
- Class/Cohort Analytics: The system shall provide an aggregate report of flag rates and risk-score distribution across an exam's roster, to help instructors identify systemic issues (e.g., a miscalibrated policy).
- Appeals Package Export: The system shall support a bundled export (recording reference, event log, AI summary, and reviewer notes) for use in formal appeals processes.
- Audit/Compliance Report: The system shall provide a report of access logs and retention actions suitable for compliance review under FERPA/GDPR (see 3.5, 3.6).

3.2 Usability

- The proctor dashboard shall present risk scores and flags in a clear, prioritized view so a single proctor can effectively monitor many simultaneous test-takers.
- Test-takers shall receive a clear, low-friction calibration/setup flow (camera/mic check, baseline capture) before the exam begins.
- Flagged events shall include enough context (timestamp, snippet/thumbnail, signal type) for a proctor to make a quick, informed judgment without replaying the full session.
- The AI summary panel shall be visually distinguishable from raw/system-generated flags so proctors do not mistake AI-generated narrative for verified fact.

3.3 Reliability & Availability

- The system shall maintain continuous monitoring for the full duration of an exam session without dropping frames/audio for extended periods; transient interruptions shall be logged and surfaced to the proctor.
- The platform shall be available with uptime sufficient to support scheduled high-stakes exam windows, with failover or graceful degradation (e.g., falling back to recording-only mode) if a realtime ML service becomes unavailable.
- If the LLM summarization service is unavailable, the dashboard shall continue to function normally using raw flags/recordings; summary generation shall retry or be requestable later without blocking proctor review.

3.4 Performance

- Head-pose, gaze, and object/person detection shall run at a frame rate sufficient for real-time flagging (e.g., near-continuous analysis rather than large sampling gaps).
- Risk scores on the proctor dashboard shall update with minimal latency relative to the underlying event.

- The system shall scale to support many concurrent exam sessions without materially degrading persession detection accuracy or dashboard responsiveness.
- On-demand session summaries shall be generated within a proctor-reasonable wait time (e.g., seconds, not minutes) for a typical session's event volume.

3.5 Security

- Data Encryption: All session recordings and biometric data (facial landmarks, gaze data, voice data) shall be encrypted in transit and at rest.
- Access Control & Audit Trails: Access to recordings, flagged events, and AI-generated summaries shall be role-restricted (e.g., proctor, administrator, appeals reviewer), with an audit trail of who viewed or acted on what data.
- LLM Data Minimization: Only structured, non-biometric event metadata shall be transmitted to the LLM summarization service; raw media and biometric feature vectors are excluded by design (see 3.1.5).

3.6 Design Constraints

- Privacy & Regulatory Compliance: The system shall support consent management, defined data retention periods, and alignment with FERPA/GDPR requirements, including special handling rules for biometric data.
- Human-in-the-Loop Requirement: The system's architecture shall enforce that no automated flag — and no LLM-generated summary — results in a final cheating determination without human proctor review. This is a hard design constraint, not merely a configurable setting.
- AI Summary Traceability: Every statement in a generated summary shall be traceable back to specific rows in the Events/Flags table, so proctors and appeals reviewers can verify the summary against underlying evidence.

3.7 Interfaces

- Test-Taker Client Interface: Captures webcam, microphone, and browser behavioral events during the exam; handles calibration and consent capture.
- Proctor Dashboard Interface: Displays real-time risk scores, flagged events, session recordings, and AI-generated summaries for review across multiple concurrent test-takers.
- Exam Platform / LMS Integration Interface: Exchanges exam session metadata (student ID, exam start/end times) with the institution's existing exam delivery or learning management system. • Database/Storage Interface: Defines the contract between the stream processor and the Events/Flags, RiskScores, Recordings, and SessionSummaries tables, plus the encrypted object storage backend.
- LLM Summarization Service Interface: Defines the request/response contract between the Events/Flags database and the LLM API — specifically, which fields are sent (student_id, timestamp,

signal_type/error_type, confidence, duration) and the expected structured, chronologically-ordered summary response format with a timestamp retained for each referenced event.